

The Basic Principles/Tools of Finance

Topics Covered

- Over a Century of Capital Market History
- Measuring Portfolio Risk
- Calculating Portfolio Risk
- Beta and Unique Risk
- Diversification & Value Additivity

Present Value Part 1

- Finance
 - Studies how people make decisions:
 - Allocation of resources over time
 - Handling of risk
- Present value
 - Amount of money today
 - That would be needed
 - Using prevailing interest rates
 - To produce a given future amount of money

Present Value, Part 2

- Future value
 - Amount of money in the future
 - That an amount of money today will yield
 - Given prevailing interest rates
- Compounding
 - Accumulation of a sum of money
 - Interest earned remains in the account
 - To earn additional interest in the future

Present Value, Part 3

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- Present value = \$100
 - Interest rate = r
 - Future value = ...
 - $(1+r)^1 \times \$100$ after 1 year,
 - $(1+r)^1 \times (1+r)^1 \times \$100 = (1+r)^2 \times \$100$ after 2 years,
 - $(1+r)^3 \times \$100$ after 3 years, ...
 - $(1+r)^N \times \$100$ after N years,

Present Value, Part 4

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- Future value = \$200 in N years
 - Interest rate = r
 - Present value = $\$200/(1+r)^N$
 - Discounting
 - Find present value for a future sum o money

Present Value, Part 5

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- General formula for discounting:
 - r , interest rate
 - X , amount to be received in N years (future value)

$$\text{Present value} = X / (1+r)^N$$

Managing Risk

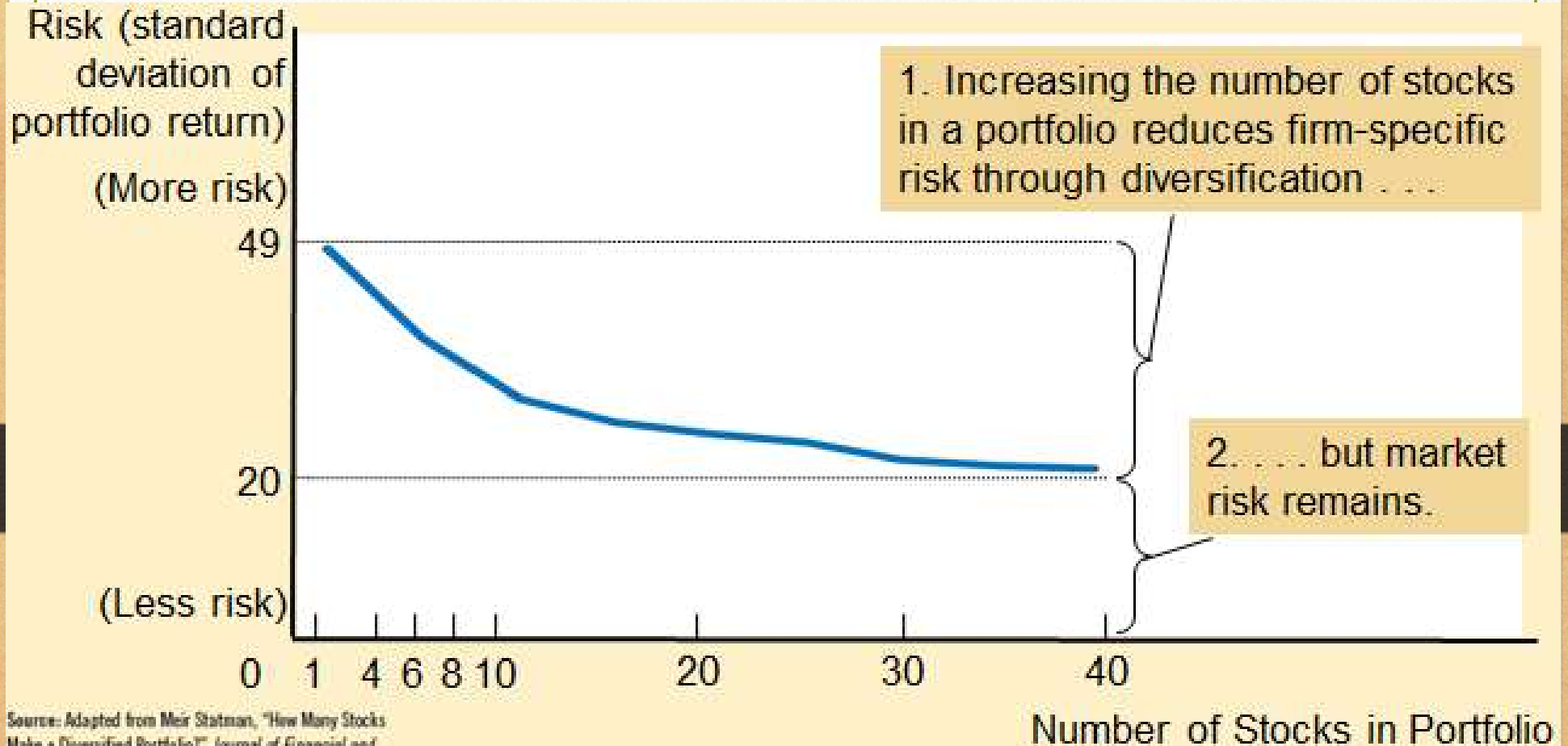
- Diversification
 - Reduction of risk
 - By replacing a single risk with a large number of smaller, unrelated risks
 - “Don’t put all your eggs in one basket”
- Risk
 - Standard deviation - measures the volatility of a variable

Managing Risk

- Risk of a portfolio of stocks
 - Depends on number of stocks in the portfolio
 - The higher the standard deviation
 - The riskier the portfolio

Figure 2

Diversification Reduces Risk



Source: Adapted from Meir Statman, "How Many Stocks Make a Diversified Portfolio?" *Journal of Financial and Quantitative Analysis* 22 (September 1987): 353-364.

- This figure shows how the risk of a portfolio, measured here with a statistic called the standard deviation, depends on the number of stocks in the portfolio. The investor is assumed to put an equal percentage of her portfolio in each of the stocks. Increasing the number of stocks reduces, but does not eliminate, the amount of risk in a stock portfolio.

Managing Risk, Part 7

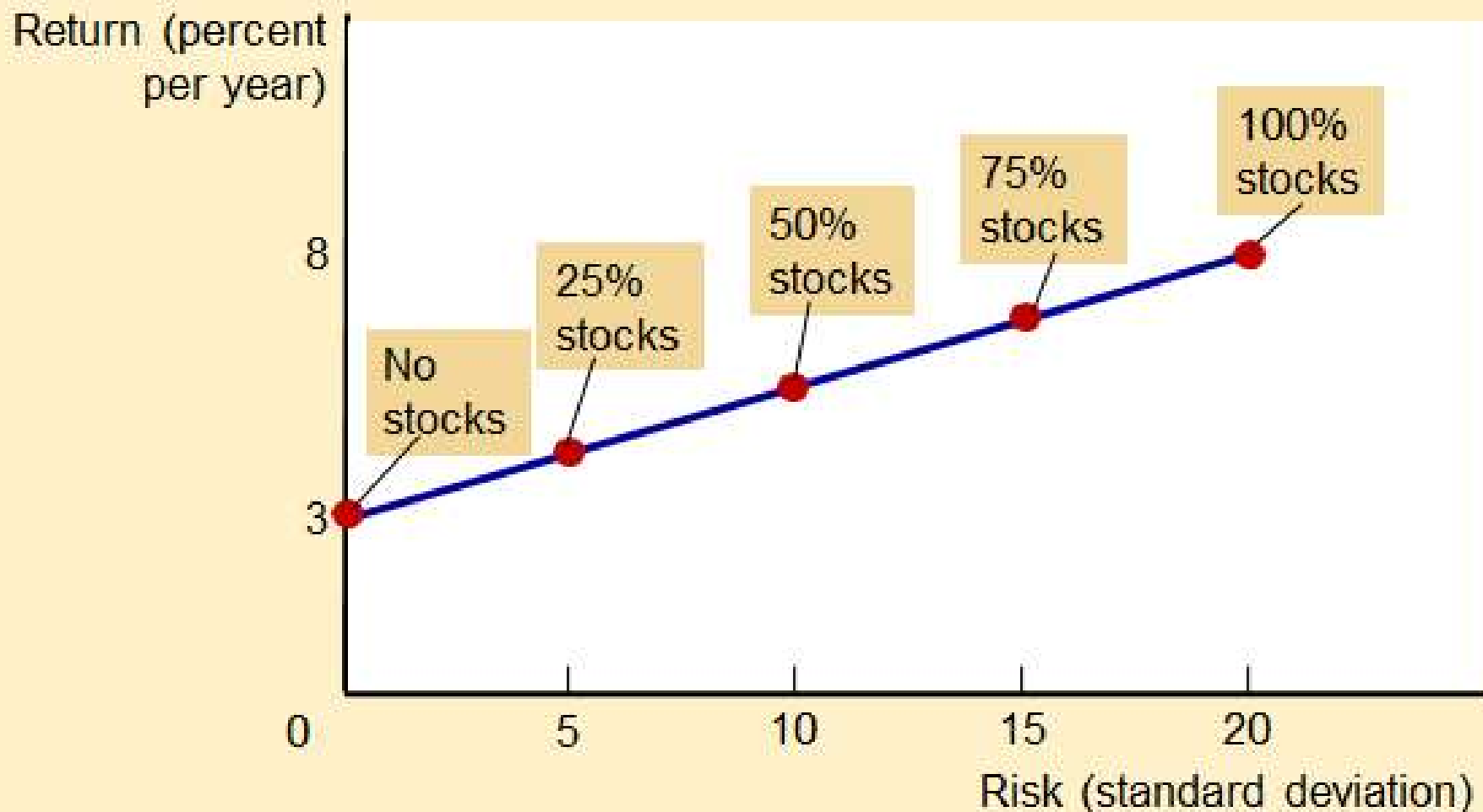
- Diversification
 - Can eliminate firm-specific risk
 - Cannot eliminate market risk
- Firm-specific risk
 - Affects only a single company
- Market risk
 - Affects all companies in the stock market

Managing Risk, Part 8

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- The trade-off between risk and return
 - Two types of assets
 - Diversified group
 - 8% return
 - 20% standard deviation
 - Safe alternative
 - 3% return
 - 0% standard deviation
 - The more a person puts into stocks
 - The greater the risk and the return

Figure 3

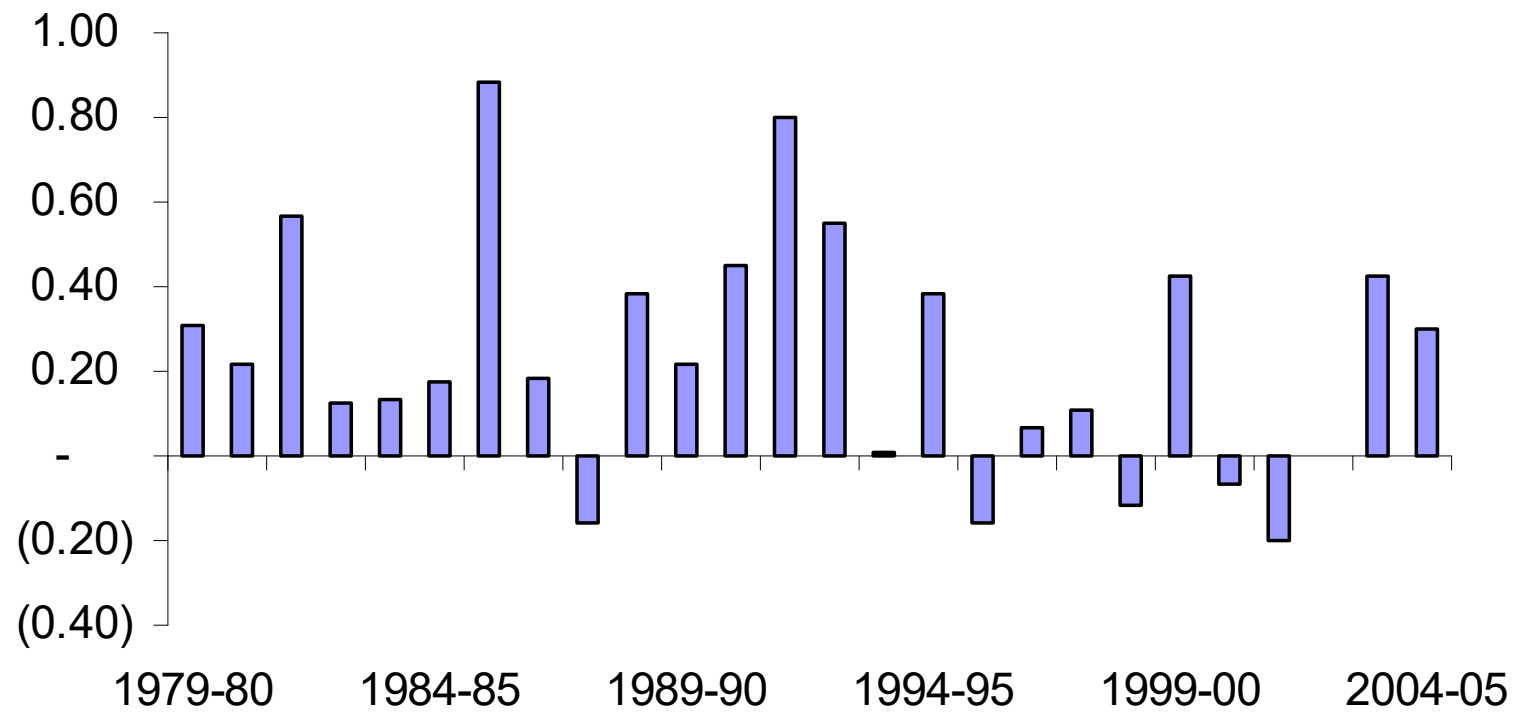
The Trade-off between Risk and Return



- When people increase the percentage of their savings that they have invested in stocks, they increase the average return they can expect to earn, but they also increase the risks they face.

Stock Index Returns in India

Rate of return, percent



Measuring Risk

Variance - Average value of squared deviations from mean. A measure of volatility.

Standard Deviation - Average value of squared deviations from mean. A measure of volatility.

Measuring Risk

Diversification - Strategy designed to reduce risk by spreading the portfolio across many investments.

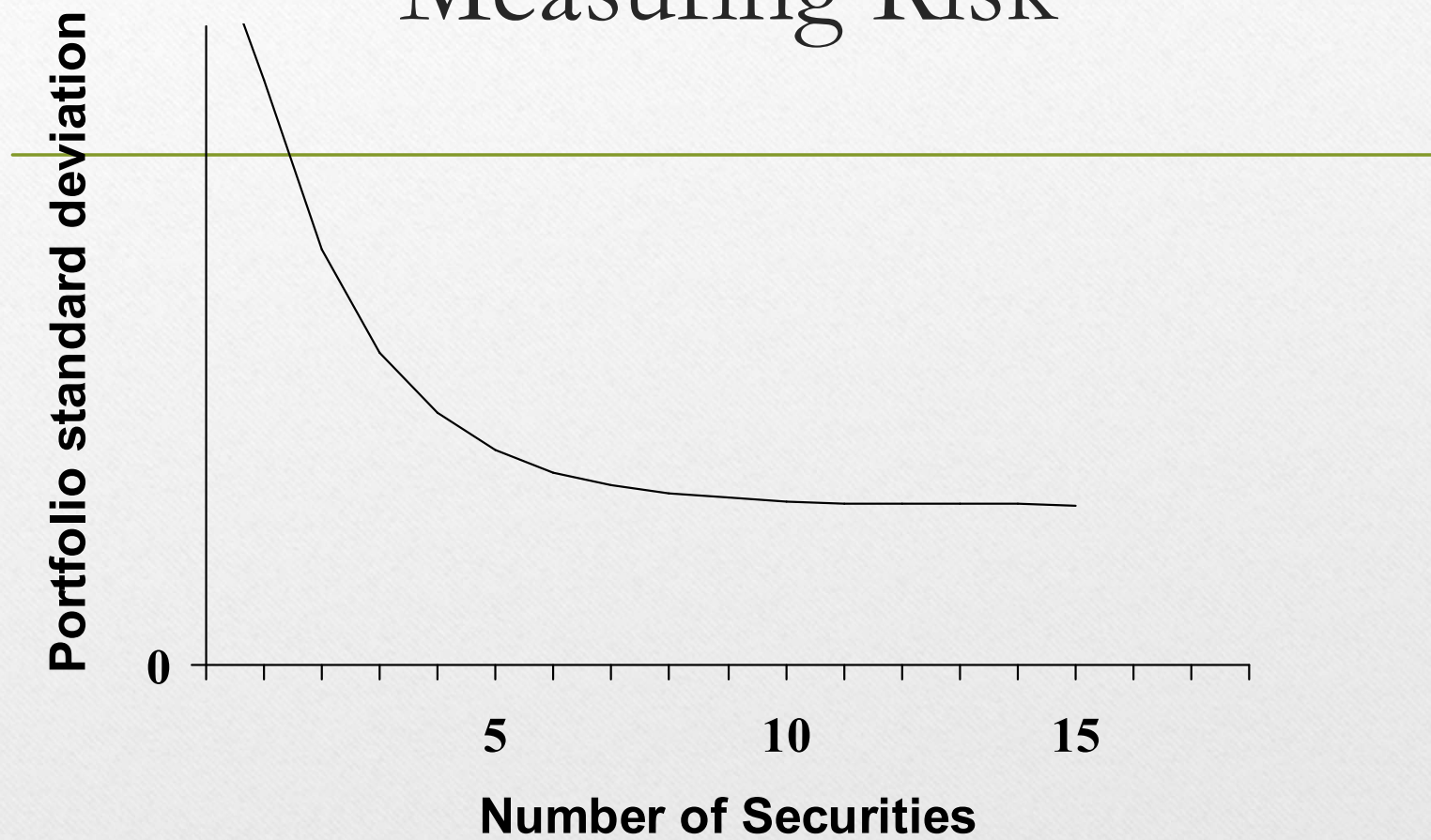
Unique Risk - Risk factors affecting only that firm.
Also called “diversifiable risk.”

Market Risk - Economy-wide sources of risk that affect the overall stock market. Also called “systematic risk.”

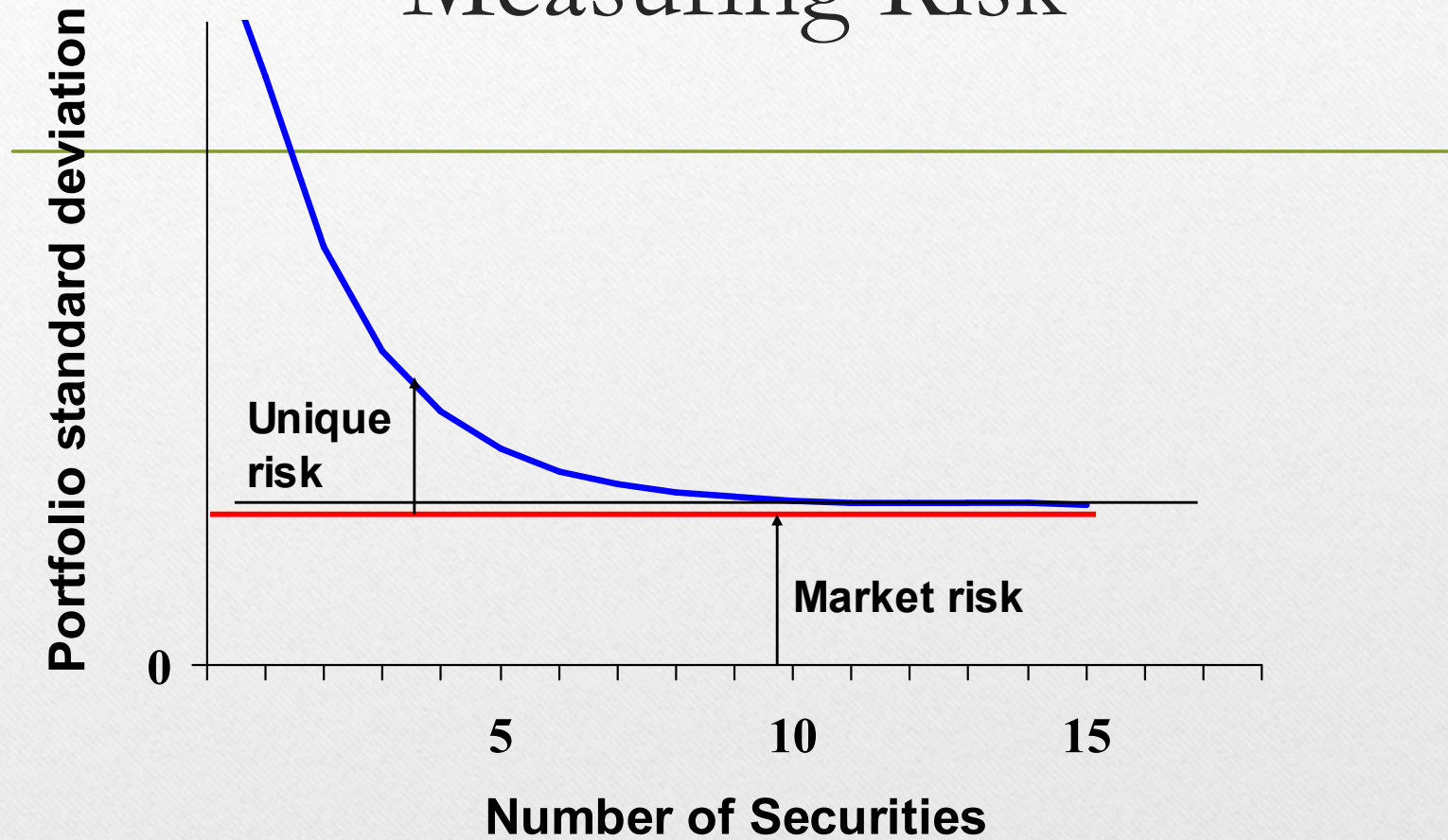
Measuring Risk

$$\begin{aligned} \text{Portfolio rate of return} &= \left(\begin{array}{c} \text{fraction of portfolio} \\ \text{in first asset} \end{array} \right) \times \left(\begin{array}{c} \text{rate of return} \\ \text{on first asset} \end{array} \right) \\ &+ \left(\begin{array}{c} \text{fraction of portfolio} \\ \text{in second asset} \end{array} \right) \times \left(\begin{array}{c} \text{rate of return} \\ \text{on second asset} \end{array} \right) \end{aligned}$$

Measuring Risk



Measuring Risk



The variance of a two stock portfolio is the sum of these four boxes

	Stock 1	Stock 2
Stock 1	$x_1^2 \sigma_1^2$	$x_1 x_2 \sigma_{12} =$ $x_1 x_2 \rho_{12} \sigma_1 \sigma_2$
Stock 2	$x_1 x_2 \sigma_{12} =$ $x_1 x_2 \rho_{12} \sigma_1 \sigma_2$	$x_2^2 \sigma_2^2$

Example Portfolio Risk

Suppose you invest 47% of your portfolio in Reliance Energy and 53% in Grasim Industries. The expected return on your Reliance Energy stock is 17% and on Grasim is 14%. The expected return on your portfolio is:

$$\text{Expected Return} = (.47 \times 17) + (.53 \times 14) = 15.41\%$$

Example

Portfolio Risk

Suppose you invest 47% of your portfolio in Reliance Energy and 53% in Grasim Industries. The expected return on your Reliance Energy stock is 17% and on Grasim is 14%. **The standard deviation of their annualized daily returns are 37% and 33%, respectively. Assume a correlation coefficient of 1.0 and calculate the portfolio variance.**

	Reliance Energy	Grasim
Reliance Energy	$x_1^2 \sigma_1^2 = (0.47)^2 \times (37)^2$	$x_1 x_2 \rho_{12} \sigma_1 \sigma_2$ $= (0.47) \times (0.53) \times 1 \times (37) \times (33)$
Grasim	$x_1 x_2 \rho_{12} \sigma_1 \sigma_2$ $= (0.47) \times (0.53) \times 1 \times (37) \times (33)$	$x_2^2 \sigma_2^2 = (0.53)^2 \times (33)^2$

Example

Portfolio Risk

Suppose you invest 47% of your portfolio in Reliance Energy and 53% in Grasim Industries. The expected return on your Reliance Energy stock is 17% and on Grasim is 14%. The standard deviation of their annualized daily returns are 37% and 33%, respectively. Assume a correlation coefficient of 1.0 and calculate the portfolio variance.

$$\begin{aligned}\text{Portfolio variance} &= [(0.47)^2 \times (37)^2] + [(0.53)^2 \times (33)^2] + 2 (0.47 \times 0.53 \times 1 \times 33 \times 37) \\ &= 1216.61\end{aligned}$$

The standard deviation is

$$\sqrt{1216.61} = 34.88 \text{ percent}$$

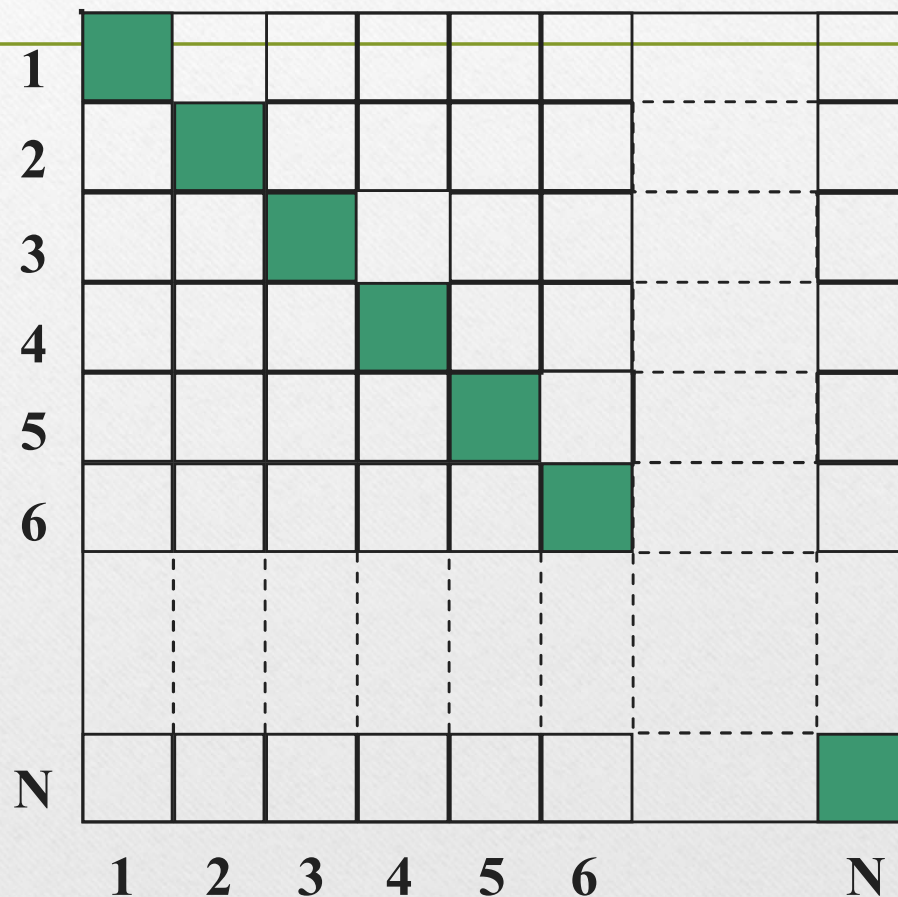
Portfolio Risk

$$\text{Expected Portfolio Return} = (x_1 r_1) + (x_2 r_2)$$

$$\text{Portfolio Variance} = x_1^2 \sigma_1^2 + x_2^2 \sigma_2^2 + 2(x_1 x_2 \rho_{12} \sigma_1 \sigma_2)$$

Portfolio Risk
The shaded boxes contain variance terms; the remainder contain covariance terms.

STOCK



To calculate
portfolio
variance add
up the boxes

STOCK

Beta and Unique Risk
Market Portfolio - Portfolio of all assets in the economy. In practice a broad stock market index, such as the Nifty or Sensex, is used to represent the market.

Beta - Sensitivity of a stock's return to the return on the market portfolio.

Beta and Unique Risk

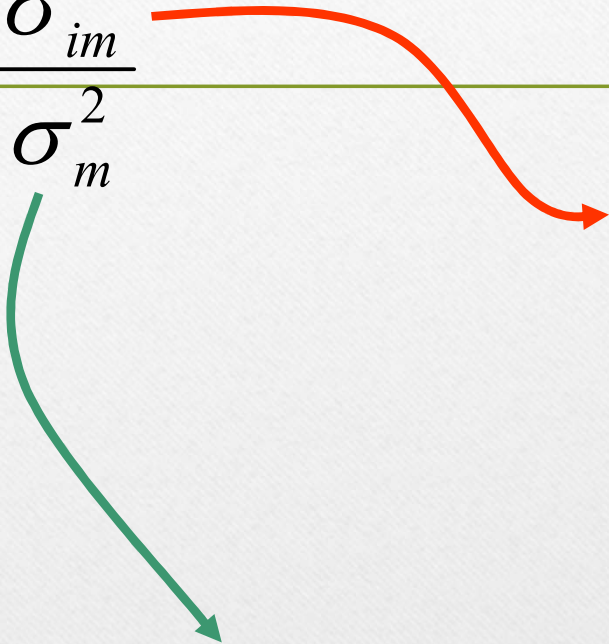
$$B_i = \frac{\sigma_{im}}{\sigma_m^2}$$

Beta and Unique Risk

$$B_i = \frac{\sigma_{im}}{\sigma_m^2}$$

Covariance with the
market

Variance of the market



Asset Valuation, Part 1

- Fundamental analysis
 - Study of a company's accounting statements and future prospects
 - To determine its value
- Undervalued stock: $\text{Price} < \text{value}$
- Overvalued stock: $\text{Price} > \text{value}$
- Fairly valued stock: $\text{Price} = \text{value}$

Asset Valuation, Part 2

- Use fundamental analysis to pick a stock
 - Do all the necessary research yourself
 - Rely on the advice of Wall Street analysts/BSE/NSE
 - Buy a mutual fund
 - A manager conducts fundamental analysis and makes the decision for you

Asset Valuation, Part 3

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- The efficient markets hypothesis
 - Asset prices reflect all publicly available information about the value of an asset
 - Each company listed on a major stock exchange is followed closely by many money managers
 - Equilibrium of supply and demand sets the market price

Asset Valuation, Part 4

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- Stock markets
 - Exhibit informational efficiency
 - Informational efficiency
 - Description of asset prices
 - Rationally reflect all available information
 - Implication of efficient markets hypothesis
 - Stock prices should follow a random walk
 - Changes in stock prices are impossible to predict from available information

Asset Valuation, Part 5

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- Efficient markets hypothesis
 - Assumes that people buying & selling stock are rational
 - Process information about stock's underlying value
 - Fluctuations in stock prices
 - Partly psychological

Asset Valuation, Part 6

- When price of an asset
 - Above its fundamental value
 - Market - experiencing a speculative bubble

- Possibility of speculative bubbles
 - Value of the stock to a stockholder depends on:
 - Stream of dividend payments
 - Final sale price

Asset Valuation, Part 7

- Debate: frequency and importance of departures from rational pricing
 - Market irrationality

 - Movement in stock market